Crooked cylinder

X19 (2019) — English translation

A cylindrical vessel of height $H = 50$ cm with mirror walls is filled with a transparent liquid, the refractive index of which decreases with height according to a linear law. Near the bottom of the cylinder there is a point light source with power $P = 100$ Вт. The refractive index of the liquid near the bottom is $n_1 = 1.50$, and at the surface $n_2 = 1.44$. The bottom of the vessel and its lid are covered with a substance that absorbs all radiation that hits it

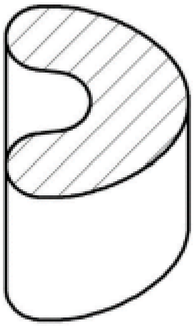

Fig. 1.

A1	What radiation power $P_д$ is absorbed by the bottom of the cylinder?	**0**
A2	Suppose a certain beam passes near the cylinder cover. What is the radius of curvature $R_{кр}$ of this ray?	**0**
A3	What is the radius of curvature $R_д$ of the same ray directly near the source?	**0**

Source: https://pho.rs/p/3059